

Structured Gaussian Mapping for Memory-Efficient Immersive XR

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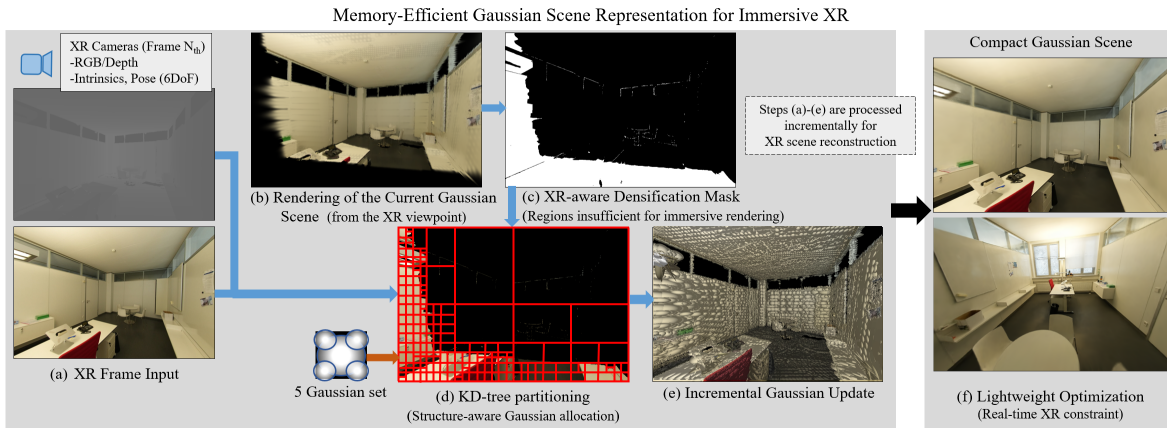


Figure 1: Overview of the proposed Structured Gaussian Mapping pipeline for immersive XR. Sequential RGB-D frames and 6DoF device poses obtained from an XR tracking and mapping system are processed incrementally. The accumulated Gaussian scene is rendered from the current XR viewpoint (b), and regions that are insufficiently represented for immersive rendering are identified via an XR-aware densification mask (c). The masked image space is partitioned using a KD-tree (d), enabling structure-aware and compact Gaussian allocation and incremental updates (e), followed by lightweight Gaussian optimization under XR-oriented efficiency constraints (f).

ABSTRACT

High-fidelity and scalable 3D scene representations are critical for immersive VR and XR systems under strict latency and memory constraints. Although recent studies demonstrate real-time 3D Gaussian Splatting on head-mounted displays, uncontrolled Gaussian growth during RGB-D mapping remains a key scalability bottleneck. We propose **Structured Gaussian Mapping**, a KD-tree guided allocation framework that limits Gaussian growth at the mapping stage for sequential RGB-D SLAM. By partitioning image space and allocating compact Gaussian sets only in perceptually relevant regions, our approach replaces pixel-wise densification. Experiments show up to **82%** fewer Gaussians with improves PSNR, enabling memory-efficient and scalable XR-ready Gaussian mapping.

Index Terms: 3D Gaussian Splatting, RGB-D SLAM, KD-tree Partitioning, Scalable XR Mapping.

1 MOTIVATION AND VR/XR RELEVANCE

3D Gaussian Splatting has recently emerged as a promising representation for immersive VR and XR, enabling real-time, photo-realistic rendering. Several studies have demonstrated Gaussian-based rendering directly on head-mounted displays using foveated or VR-specific optimizations. However, when Gaussian primitives are densely allocated during sequential RGB-D mapping, memory

usage and rendering workload grow rapidly, violating the strict latency and resource constraints of immersive systems.

Most VR-oriented approaches address this issue **at the rendering stage** through view-dependent or foveated optimizations [1, 3]. While effective in reducing runtime costs, they do not prevent uncontrolled Gaussian growth during reconstruction. We argue that the scalability bottleneck originates earlier in the pipeline **at the scene representation stage**, particularly under incremental RGB-D SLAM.

Goal. We aim to enable scalable and memory-efficient Gaussian scene representations by structurally limiting Gaussian growth during sequential RGB-D mapping, thereby complementing VR-oriented rendering optimizations and improving practical deployability in real-time immersive VR/XR pipelines.

2 STRUCTURED GAUSSIAN MAPPING

To mitigate uncontrolled Gaussian growth during sequential RGB-D mapping, we propose **Structured Gaussian Mapping**, which replaces **uniform pixel-wise densification** with a structured, deterministic allocation strategy tailored for memory-efficient immersive VR/XR pipelines.

2.1 Masking and KD-tree Partitioning

Given an accumulated Gaussian map and a new RGB-D frame, we render the scene from the current viewpoint (Fig. 1(b)) and identify insufficiently represented regions using a densification mask based on depth discrepancy and Gaussian contribution (Fig. 1(c)). The masked image space is then partitioned using a KD-tree, as illustrated in Fig. 1(d), with split decisions guided by depth variance, color variance, gradient magnitude, and Laplacian response.

2.2 Leaf-to-Gaussian Conversion

Each KD-tree leaf is converted into a compact set of Gaussian primitives. A representative Gaussian is initialized using the mean posi-

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Table 1: Gaussian reduction and rendering quality comparison.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	#Gaussians $\downarrow$
ScanNet++				
Masked Dense	25.39	0.8616	0.2650	2.42M
Ours	26.88	0.8918	0.2363	0.44M ($\downarrow 82\%$)
Hand-held RGB-D SLAM				
Masked Dense	25.88	0.8626	0.2490	7.53M
Ours	26.15	0.8706	0.2561	4.92M ($\downarrow 35\%$)

tion and color of the leaf’s points, with the scale determined by the leaf area and the orientation aligned with the estimated surface normal. For large leaves exceeding a size threshold, a small set of Gaussians (one center and four corners) is used to better approximate rectangular regions induced by the KD-tree partitioning (Fig. 1(d)).

2.3 Optimization

The resulting Gaussians are optimized using differentiable rendering with a photometric consistency loss across views after all frames have been incrementally inserted (Fig. 1(f)). At this stage, conventional densification and pruning are jointly applied to refine the scene representation. A lightweight global optimization (approximately 10 iterations) ensures fast convergence and stable refinement.

3 EXPERIMENTS

Datasets

- ScanNet++ (scene 8b5caf3398): A high-quality indoor RGB-D sequence with ground-truth camera poses [4].
- Hand-held RGB-D SLAM sequence: A large, texture-rich scene containing pose noise and motion blur.

Baseline We compare against **Masked Dense Mapping**, a common Gaussian-based SLAM strategy that allocates one Gaussian per valid pixel and relies on pruning to control growth [2].

Quantitative Results Structured Gaussian Mapping reduces Gaussian count while maintaining or improving rendering quality. On ScanNet++, our method reduces the number of Gaussians by **82%** while improving PSNR by **+1.49 dB**, demonstrating that structured allocation suppresses redundancy without sacrificing fidelity. On the hand-held SLAM sequence, which is larger and noisier, we still achieve a **35% reduction** in Gaussians with comparable or improved rendering quality, as shown in Table 1.

Qualitative Results Figure 2 visualizes the spatial distribution of Gaussian primitives produced during sequential RGB-D mapping. Masked dense mapping places Gaussians uniformly at the pixel level, resulting in redundant and spatially cluttered distributions, particularly on planar and low-texture regions. In contrast, the proposed Structured Gaussian Mapping allocates Gaussians **more compactly** by **adapting their density to scene complexity**.

Performance Experiments were conducted on an Intel® Core™ i9-12900K CPU and an NVIDIA A6000 Ada GPU. KD-tree construction is parallelized in C and runs at over **10 fps** at a resolution of 876×584 , with comparable processing cost to the non-parallelized masked dense baseline, and full optimization completing in under one minute.

4 CONCLUSION AND DISCUSSION

We presented Structured Gaussian Mapping, a KD-tree guided approach for compact Gaussian scene reconstruction from sequential RGB-D SLAM. By replacing pixel-wise densification with

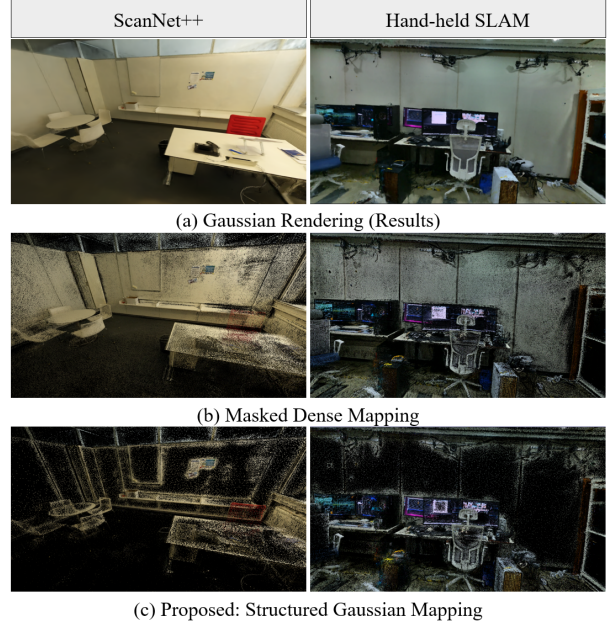


Figure 2: Visualization of Gaussian placement under sequential RGB-D mapping: (a) rendered output, (b) masked dense mapping, and (c) the proposed Structured Gaussian Mapping.

structure-aware allocation, the proposed method effectively limits Gaussian growth at the representation level.

Experiments show that our approach reduces the number of Gaussians by up to 82% while preserving or improving rendering quality, making Gaussian Splatting more practical for memory-constrained immersive VR/XR systems. Future work will extend this approach to dynamic scenes and large-scale outdoor environments.

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